a) Find the Area of a Circle Using math

```
import math

radius = float(input("Enter radius: "))
area = math.pi * radius * radius

print("Area of Circle =", area)
```

Line-wise Explanation

`import math`
Imports the math module to use mathematical constants like π (pi).
`radius = float(input("Enter radius: "))`
Takes radius from the user and converts it into a decimal number.
`area = math.pi * radius * radius`
Calculates the area using the formula:
Area = $\pi \times r^2$
`print("Area of Circle =", area)`
Displays the area.

---

b) Calculate Square Root and Power of a Number

```
import math

num = float(input("Enter a number: "))

sqrt_num = math.sqrt(num)
power_num = math.pow(num, 2)

print("Square Root =", sqrt_num)
print("Power =", power_num)
```

Line-wise Explanation

`import math`
Imports the math module.
`num = float(input("Enter a number: "))`
Gets a number from the user.
`sqrt_num = math.sqrt(num)`
Calculates the square root.
`power_num = math.pow(num, 2)`
Calculates the square (power of 2).
`print("Square Root =", sqrt_num)`
Displays square root.
`print("Power =", power_num)`
Displays power.

---

c) Find the Factorial of a Number

```
import math

num = int(input("Enter a number: "))

fact = math.factorial(num)

print("Factorial =", fact)
```

Line-wise Explanation

`import math`
Imports math module.
`num = int(input("Enter a number: "))`
Takes an integer input.
`fact = math.factorial(num)`
Calculates factorial.
`print("Factorial =", fact)`
Displays factorial value.

---

d) Generate a Random Float and Integer

```
import random

float_num = random.random()
int_num = random.randint(1, 10)

print("Random Float =", float_num)
print("Random Integer =", int_num)
```

Line-wise Explanation

`import random`
Imports random module.
`float_num = random.random()`
Generates a random decimal number between 0 and 1.

`int_num = random.randint(1, 10)`
Generates a random integer from 1 to 10.
`print("Random Float =", float_num)`
Displays float value.
`print("Random Integer =", int_num)`
Displays integer value.

---

e) Guess the Number Game (1–10)

```
import random

secret = random.randint(1, 10)

guess = int(input("Guess a number (1-10): "))

if guess == secret:
    print("Correct!")
else:
    print("Wrong! Number was", secret)
```

Line-wise Explanation
`import random`
Imports random module.
`secret = random.randint(1, 10)`
Generates a secret number.
`guess = int(input("Guess a number (1-10): "))`
Gets user's guess.
`if guess == secret:`
Checks if guess is correct.
`print("Correct!")`
Displays success message.
`else:`
Runs if guess is wrong.
`print("Wrong! Number was", secret)`
Displays the correct number.

---

f) Choose a Team Leader Randomly

```
import random

students = ["Ram", "Sita", "Hari", "Gita"]

leader = random.choice(students)

print("Team Leader =", leader)
```

Line-wise Explanation
`students = ["Ram", "Sita", "Hari", "Gita"]`
Creates a list of students.
`leader = random.choice(students)`
Selects one name randomly.
`print("Team Leader =", leader)`
Displays selected leader.

---

g) Shuffle a Simple Deck of Cards

```
import random

cards = ["A", "K", "Q", "J", "10"]

random.shuffle(cards)

print(cards)
```

Line-wise Explanation
`cards = ["A", "K", "Q", "J", "10"]`
Creates a list of cards.
`random.shuffle(cards)`
Randomly rearranges the cards.
`print(cards)`
Displays shuffled cards.

---

h) Read a CSV and Display First 3 Rows

```
import pandas as pd

data = pd.read_csv("students.csv")

print(data.head(3))
```

Line-wise Explanation
`import pandas as pd`

Imports pandas library.
`data = pd.read_csv("students.csv")`
Reads CSV file.
`print(data.head(3))`
Displays first 3 rows.

---

i) Create a DataFrame and Show Basic Statistics

```
import pandas as pd

data = {
    "Marks": [60, 70, 80, 90, 100]
}

df = pd.DataFrame(data)

print(df.describe())
```

Line-wise Explanation
`data = {"Marks":[60,70,80,90,100]}`
Creates a dictionary.
`df = pd.DataFrame(data)`
Converts dictionary into DataFrame.
`print(df.describe())`
Shows count, mean, min, max, etc.

---

j) Draw a Simple Line Plot

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]
y = [10, 20, 15, 25, 30]

plt.plot(x, y)

plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.title("Line Plot")
plt.grid()

plt.show()
```

Line-wise Explanation
`import matplotlib.pyplot as plt`
Imports plotting library.
`x = [1,2,3,4,5]`
Stores X-axis values.
`y = [10,20,15,25,30]`
Stores Y-axis values.
`plt.plot(x, y)`
Draws line graph.
`plt.xlabel("X-axis")`
Labels X-axis.
`plt.ylabel("Y-axis")`
Labels Y-axis.
`plt.title("Line Plot")`
Adds title.
`plt.grid()`
Displays grid.
`plt.show()`
Shows graph.

---

k) Draw a Histogram

```
import matplotlib.pyplot as plt

marks = [45, 50, 60, 65, 70, 75, 80, 85, 90]

plt.hist(marks)

plt.title("Histogram")
plt.show()
```

Line-wise Explanation
`marks = [...]`
Stores marks data.
`plt.hist(marks)`
Creates histogram.
`plt.title("Histogram")`
Adds title.
`plt.show()`
Displays graph.

---

l) Create a Pie Chart

```
import matplotlib.pyplot as plt

subjects = ["Math", "Science", "English"]
marks = [40, 35, 25]

plt.pie(marks, labels=subjects)

plt.title("Pie Chart")
plt.show()
```

Line-wise Explanation

`subjects = [...]`
Stores subject names.
`marks = [...]`
Stores marks.
`plt.pie(marks, labels=subjects)`
Creates pie chart.
`plt.title("Pie Chart")`
Adds title.
`plt.show()`
Displays chart.

---

m) Create a Bar Diagram

```
import matplotlib.pyplot as plt

subjects = ["Math", "Science", "English"]
marks = [80, 70, 90]

plt.bar(subjects, marks)

plt.xlabel("Subjects")
plt.ylabel("Marks")
plt.title("Bar Diagram")

plt.show()
```

Line-wise Explanation

`plt.bar(subjects, marks)`
Creates bar chart.
`plt.xlabel("Subjects")`
Labels X-axis.
`plt.ylabel("Marks")`
Labels Y-axis.
`plt.title("Bar Diagram")`
Adds title.
`plt.show()`
Displays bar diagram.